

Build a Data Mart in SQL

Phase 1 – Conception Phase

Course: DLBDSPBDM01

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Project Overview (Adjusted)

1. Requirements Analysis

The design process began with analyzing platform requirements, focusing on three main roles: Guests, Hosts, and TravelAdmins. Each role's actions and data needs were identified. The system was adjusted based on tutor feedback to clarify who can write reviews and to improve the realism of the payment flow.

Roles and Actions:

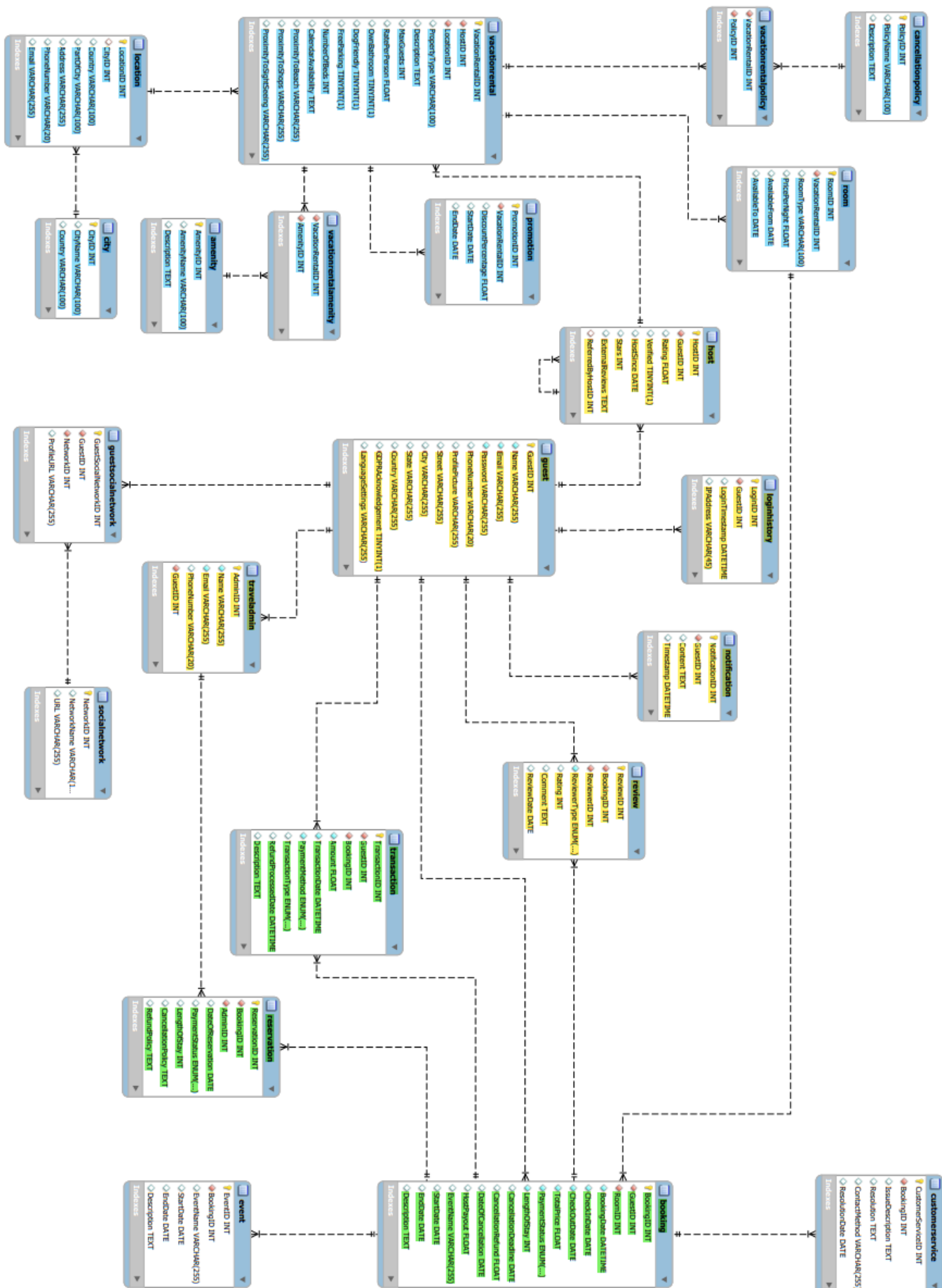
- **Guests:** Search for properties, book rentals, leave reviews, and manage profiles. The database handles these tasks by organizing and retrieving the necessary data effectively.
- **Hosts:** List properties, manage bookings, respond to guest queries, and leave reviews. Property details and availability are tracked in the database.
- **TravelAdmins:** Oversee the platform's operations, manage user accounts, handle customer service, and generate reports. Access to all stored data is required for reporting and operations.

Data and Functional Requirements:

Payment processing and transactions are now integrated, presenting a more realistic payment management. Reviews can now be written by both guests and hosts, providing a more accurate representation of how platform entities interact with each other. The database tracks users, listings, reservations, payments, and reviews, ensuring secure and reliable data handling.

2. Entity Relationship Model (ERM)

The revised ERM reflects an improved payment process, with attributes like TransactionType, PaymentMethod, and RefundProcessedDate for the transaction management. The Review entity allows both guests and hosts to leave reviews, with attributes like ReviewerType and ReviewerID. The model minimizes redundancy and ensures data integrity across the platform's 22 entities, covering profiles, reservations, payments, and reviews.



- **Triple Relationships:** Exist between tables like Guest, Booking, and Transaction to connect users, reservations, and payments. A similar structure connects Guest, Booking, and Review.
- **Recursive Relationships:** The Host table allows tracking referrals.
- **Cardinality:** Martin notation, also known as Crow's Foot notation, is used to define relationships and cardinality between entities.

Normalization:

During the design process, normalization was prioritized to minimize redundancy and enhance data integrity. For example, the City entity collects city information, while full address details (street, state, country) are stored in appropriate entities. Foreign keys link tables as Location and VacationRental to maintain consistency without repeating data.

3. Tools and Documentation

MySQL Workbench was used for ERM creation, testing, and managing the design process. Each of the 22 entities is populated with 20 entries. The SQL code documentation contains information about the database structure and covers table creation, relationships, and sample data entries. The design's functionality was confirmed through test cases that verify the proper handling of bookings and payments.

4. Data Dictionary

1. Table: Guest

Column Name	Data Type	Constraints	Description
GuestID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for guest
Name	VARCHAR(255)	NOT NULL	Guest's full name
Email	VARCHAR(255)	NOT NULL, UNIQUE	Guest's email address
Password	VARCHAR(255)	NOT NULL	Password for guest login
PhoneNumber	VARCHAR(20)		Guest's phone number
ProfilePicture	VARCHAR(255)		Guest's profile picture
Street	VARCHAR(255)		Guest's street address

City	VARCHAR(255)		City of guest
State	VARCHAR(255)		State or province of guest
Country	VARCHAR(255)		Country of guest
GDPAcknowledgement	BOOLEAN		If guest acknowledges GDPR terms
LanguageSettings	VARCHAR(255)		Guest's language

2. Table: Host

Column Name	Data Type	Constraints	Description
HostID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for host
GuestID	INT	FOREIGN KEY (GuestID)	Reference to guest who is a host
Rating	FLOAT		Host's rating
Verified	BOOLEAN		If the host is verified
HostSince	DATE		Date since they are a host
Stars	INT		Number of stars rating for host
ExternalReviews	TEXT		External reviews for the host
ReferredByHostID	INT	FOREIGN KEY (HostID)	Recursive reference to referring host

3. Table: TravelAdmin

Column Name	Data Type	Constraints	Description
AdminID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier travel admin
Name	VARCHAR(255)	NOT NULL	Admin's full name
Email	VARCHAR(255)	NOT NULL, UNIQUE	Admin's email address
PhoneNumber	VARCHAR(20)		Admin's phone number
GuestID	INT	FOREIGN KEY (GuestID)	Reference to guest

4. Table: SocialNetwork

Column Name	Data Type	Constraints	Description
NetworkID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for social network

NetworkName	VARCHAR(100)		Name of social network
URL	VARCHAR(255)		URL of the social network

5. Table: GuestSocialNetwork

Column Name	Data Type	Constraints	Description
GuestSocialNetworkID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for guest social network
GuestID	INT	FOREIGN KEY (GuestID)	Reference to guest
NetworkID	INT	FOREIGN KEY (NetworkID)	Reference to social network
ProfileURL	VARCHAR(255)		URL to guest's social profile

6. Table: LoginHistory

Column Name	Data Type	Constraints	Description
LoginID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for login history
GuestID	INT	FOREIGN KEY (GuestID)	Reference to guest
LoginTimestamp	DATETIME		Timestamp of guest login
IPAddress	VARCHAR(45)		IP address used during login

7. Table: Notification

Column Name	Data Type	Constraints	Description
NotificationID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for notification
GuestID	INT	FOREIGN KEY (GuestID)	Reference to guest
Content	TEXT		Content of the notification
Timestamp	DATETIME		Time of the notification

8. Table: City

Column Name	Data Type	Constraints	Description
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CityID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for city
CityName	VARCHAR(100)		Name of the city
Country	VARCHAR(100)		Country of the city

9. Table: Location

Column Name	Data Type	Constraints	Description
LocationID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for location
CityID	INT	FOREIGN KEY (CityID)	Reference to city
Country	VARCHAR(100)		Country of the location
PartOfCity	VARCHAR(100)		Sub-part of the city
Address	VARCHAR(255)		Full address of the location
PhoneNumber	VARCHAR(20)		Contact phone number
Email	VARCHAR(255)		Contact email address

10. Table: VacationRental

Column Name	Data Type	Constraints	Description
VacationRentalID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for vacation rental
HostID	INT	FOREIGN KEY (HostID)	Reference to host
LocationID	INT	FOREIGN KEY (LocationID)	Reference to location
PropertyType	VARCHAR(100)		Type of property
Description	TEXT		Description of vacation rental
MaxGuests	INT		Maximum guests allowed
RatePerPerson	FLOAT		Rate charged per person
OwnBathroom	BOOLEAN		If rental has its own bathroom
DogFriendly	BOOLEAN		If rental is dog-friendly
FreeParking	BOOLEAN		If free parking is available
NumberOfBeds	INT		Number of beds in rental
CalendarAvailability	TEXT		Calendar of rental availability

ProximityToBeach	VARCHAR(255)		Distance from the beach
ProximityToShops	VARCHAR(255)		Distance from shops
ProximityToSightSeeing	VARCHAR(255)		Distance from sightseeing spots

11. Table: Room

Column Name	Data Type	Constraints	Description
RoomID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for room
VacationRentalID	INT	FOREIGN KEY (VacationRentalID)	Reference to vacation rental
RoomType	VARCHAR(100)		Type of room
PricePerNight	FLOAT		Price charged per night
AvailableFrom	DATE		Availability start date
AvailableTo	DATE		Availability end date

12. Table: Amenity

Column Name	Data Type	Constraints	Description
AmenityID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for amenity
AmenityName	VARCHAR(100)		Name of the amenity
Description	TEXT		Description of the amenity

13. Table: VacationRentalAmenity

Column Name	Data Type	Constraints	Description
VacationRentalID	INT	FOREIGN KEY (VacationRentalID)	Reference to vacation rental
AmenityID	INT	FOREIGN KEY (AmenityID)	Reference to amenity
PRIMARY KEY (VacationRentalID, AmenityID)	Compound primary key (using both columns)		

14. Table: CancellationPolicy

Column Name	Data Type	Constraints	Description
PolicyID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for cancellation policy
PolicyName	VARCHAR(100)		Name of the policy
Description	TEXT		Description of the policy

15. Table: VacationRentalPolicy

Column Name	Data Type	Constraints	Description
VacationRentalID	INT	FOREIGN KEY (VacationRentalID)	Reference to vacation rental
PolicyID	INT	FOREIGN KEY (PolicyID)	Reference to cancellation policy
PRIMARY KEY (VacationRentalID, PolicyID)	Compound primary key using both columns		

16. Table: Booking

Column Name	Data Type	Constraints	Description
BookingID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for booking
GuestID	INT	FOREIGN KEY (GuestID)	Reference to guest
RoomID	INT	FOREIGN KEY (RoomID)	Reference to room
BookingDate	DATETIME	NOT NULL	Date and time of booking
CheckInDate	DATE	NOT NULL	Guest check-in date
CheckOutDate	DATE	NOT NULL	Guest check-out date
TotalPrice	FLOAT	NOT NULL	Total price for the booking
PaymentStatus	ENUM	('Paid', 'Pending', 'Cancelled')	Payment status
LengthOfStay	INT	NOT NULL	Duration of stay in days
CancellationDeadline	DATE		Deadline to cancel without penalty

CancellationRefund	FLOAT		Refund amount if cancelled
DateOfCancellation	DATE		Date when booking was cancelled
HostPayout	FLOAT		Amount paid to host
EventName	VARCHAR(255)		Name of any associated event
StartDate	DATE		Start date of any associated event
EndDate	DAT		End date of any associated event
Description	TEXT		Additional description

17. Table: Transaction

Column Name	Data Type	Constraints	Description
TransactionID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for transaction
GuestID	INT	FOREIGN KEY (GuestID)	Reference to guest
BookingID	INT	FOREIGN KEY (BookingID)	Reference to booking
Amount	FLOAT	NOT NULL	Transaction amount
TransactionDate	DATETIME	NOT NULL	Date and time of the transaction
PaymentMethod	ENUM	('CreditCard', 'BankTransfer', etc.)	Payment method used
TransactionType	ENUM	('Payment', 'Refund')	If transaction is a payment or refund
RefundProcessedDate	DATETIME		Date if refund was processed
Description	TEXT	NOT NULL	Description of the transaction (reason)

18. Table: Reservation

Column Name	Data Type	Constraints	Description
ReservationID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for reservation

BookingID	INT	FOREIGN KEY (BookingID)	Reference to booking
AdminID	INT	FOREIGN KEY (AdminID)	Reference to travel admin
DateOfReservation	DATE		Date the reservation was made
PaymentStatus	ENUM	('Paid', 'Pending', 'Cancelled')	Payment status
LengthOfStay	INT		Duration of stay in days
CancellationPolicy	TEXT		Cancellation policy details
RefundPolicy	TEXT		Refund policy details

19. Table: Review

Column Name	Data Type	Constraints	Description
ReviewID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for review
BookingID	INT	FOREIGN KEY (BookingID)	Reference to booking
ReviewerID	INT	FOREIGN KEY (GuestID)	Reference to guest/host giving review
ReviewerType	ENUM	('Guest', 'Host')	If review is from guest or host
Rating	INT		Rating provided in the review
Comment	TEXT		Review comments
ReviewDate	DATE		Date the review was posted

20. Table: CustomerService

Column Name	Data Type	Constraints	Description
CustomerServiceID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for customer service issue
BookingID	INT	FOREIGN KEY (BookingID)	Reference to booking
IssueDescription	TEXT		Description of customer issue
Resolution	TEXT		Resolution to the issue
ContactMethod	VARCHAR(255)		Contact method used to address issue
ResolutionDate	DATE		Date issue was resolved

21. Table: Event

Column Name	Data Type	Constraints	Description
EventID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for event
BookingID	INT	FOREIGN KEY (BookingID)	Reference to booking
EventName	VARCHAR(255)		Name of the event
StartDate	DATE		Event start date
EndDate	DATE		Event end date
Description	TEXT		Description of the event

22. Table: Promotion

Column Name	Data Type	Constraints	Description
PromotionID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for promotion
VacationRentalID	INT	FOREIGN KEY (VacationRentalID)	Reference to vacation rental
DiscountPercentage	FLOAT		Percentage discount offered
StartDate	DATE		Promotion start date
EndDate	DATE		Promotion end date